

Node.js Installation and First Project Setup –

1. Overview of Required Tools

To start working with Node.js, the following tools are required:

1. **Node.js** – One-time installation (runtime environment + npm).
2. **Code Editor** – Visual Studio Code (VS Code), a free editor.
3. **Express.js Project** – Installed separately for each project (covered in the next video).

This tutorial covers **Node.js and VS Code installation** and creating a **basic Node.js project**.

2. Downloading Node.js

Steps:

1. Open a web browser and go to **google.com**.
 2. Search for **Node.js**.
 3. Open the official website: **nodejs.org**.
 4. Download the Node.js setup file suitable for your operating system.
 5. Save the installer file to your system.
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3. Installing Node.js

Installation Steps:

1. Double-click the downloaded Node.js installer.
2. Click **Next**.
3. Accept the license agreement and click **Next**.
4. Choose the installation location (default: **C:\Program Files\nodejs**).
5. Continue with **Next**.

What Node.js Installer Sets Up:

- Node.js runtime environment
 - npm (Node Package Manager)
 - Automatically configures **Environment Variables (PATH)**
6. Click **Install** and allow permissions if prompted.
 7. Once installation completes, click **Finish**.

Optional: If extra tools were selected during installation, PowerShell may open and complete additional setup automatically.

4. Verifying Node.js and npm Installation

Steps:

1. Open **Command Prompt (CMD)**.
2. Run the following commands:

```
node -v
```

- ✓ Displays Node.js version if installed correctly.

```
npm -v
```

- ✓ Displays npm version.

If version numbers appear, Node.js and npm are installed successfully.

5. Fixing Environment Variable (PATH) Issues (If Node Is Not Recognized)

Steps:

1. Search for **Edit the system environment variables**.
2. Open **System Properties** → **Environment Variables**.
3. Under **System Variables**, find and double-click **Path**.
4. Ensure the Node.js path exists, for example:

```
C:\Program Files\nodejs\
```

5. If missing, add it manually.
6. Click **OK** on all windows.
7. Restart Command Prompt and recheck:

```
node -v
```

6. Installing Visual Studio Code (VS Code)

Steps:

1. Download VS Code from its official website.

2. Run the installer.
3. Accept the license agreement.
4. Click **Next** through default settings.
5. Enable the following options:
 - Open with Code (right-click menu)
 - Add to PATH
 - Desktop icon (optional)
6. Click **Install**.

Once installed, open **Visual Studio Code** from the Start Menu.

7. Recommended VS Code Extensions for Node.js & Express

Extensions are optional but improve productivity.

Suggested Extensions:

- **Express JS Snippets**
- **Express JS Snippets (alternate versions)**
- **EJS Language Support** (for template engine usage)
- **MongoDB for VS Code**
- **Path IntelliSense**
- **ESLint** (JavaScript error detection)

Benefits:

- Auto-generate large code blocks using short commands (snippets)
 - Faster coding
 - Early error detection
 - Better project navigation
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8. Creating a New Node.js Project

Step 1: Create and Open a Project Folder

1. Open VS Code.
 2. Click **Open Folder**.
 3. Create a new folder (e.g., `first-node-project`).
 4. Open and trust the folder.
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9. Initializing Node.js Project (`package.json`)

Open Terminal in VS Code:

- Menu → **Terminal** → **New Terminal**

Run Command:

```
npm init -y
```

Purpose:

- Initializes a new Node.js project
- Automatically creates `package.json`
- Skips manual questions using `-y`

`package.json` Contains:

- Project name
- Version
- Entry file (`index.js`)
- Author
- Scripts
- Dependencies (installed packages)

`package.json` is mandatory for any Node.js project.

10. Fixing PowerShell Execution Policy Error (If Occurs)

Error:

PowerShell blocks npm commands due to security restrictions.

Solution:

1. Check current policy:

```
Get-ExecutionPolicy
```

2. Set policy for current user:

```
Set-ExecutionPolicy RemoteSigned -Scope CurrentUser
```

3. Verify:

```
Get-ExecutionPolicy -List
```

After this, npm commands will work correctly.

11. Creating and Running First Node.js File

Step 1: Create File

Create a file named:

```
index.js
```

Step 2: Add JavaScript Code

```
console.log("Hello");
```

Step 3: Run the File

In terminal:

```
node index.js
```

✓ Output:

```
Hello
```

You can also run:

```
node index
```

12. Testing Code Changes

Modify the file:

```
console.log("Hello World");
```

Save the file and rerun:

```
node index.js
```

✓ Output updates accordingly.
